6.6.2 Show nth Fourier coefficient of $g_2^{\bar{v}}(\tau)$ is bounded by Cn^2 for some constant C.

Pf: We want to show $-\frac{\pi}{N^2 \operatorname{Im}(\tau)}$ and $\frac{C_2}{N^2}\sigma(n)$ can be controled by some Cn^2.

Essentially,
$\sigma(n) \sim n$.

$$\sigma(n) = \sum_{0<d|n} d \overset{n \text{ even } n=2n'}{\leq} 1+\dots+n' + n = \frac{\frac{n}{2}(1+\frac{n}{2})}{2} + n = \frac{n^2+2n}{8} + n \sim n^2.$$

6.6.3 Show $G_{2,\mathbb{1}_1,\mathbb{1}_1}(\tau) - tG_{2,\mathbb{1}_1,\mathbb{1}_1}(t\tau) = \frac{C_2}{N^2}\left(E_{2,\mathbb{1}_1,\mathbb{1}_1}(\tau) - tE_{2,\mathbb{1}_1,\mathbb{1}_1}(t\tau)\right)$

$$= \frac{1}{N^2} G_{2,t}(\tau).$$

Pf: $g(\bar{\mathbb{1}}_1)=1$, $v=N$ gives first identity by $G_2^{\psi,\varphi}(\tau) = \frac{C_2 g(\bar{\varphi})}{v^2} E_2^{\psi,\varphi}(\tau)$.

$$G_{2,t}(\tau) = G_2(\tau) - tG_2(t\tau) = \underline{2\zeta(2)}\left(E_2(\tau) - tE_2(t\tau)\right)$$

$$G_{2,\mathbb{1}_1,\mathbb{1}_1}(\tau) = \sum_{c=0}\sum_{d=0}\sum_{e=0} G_2^{(c,d+e)}(\tau) = G_2^{\overline{(0,0)}}(\tau) = \frac{1}{N^2} G_2(\tau).$$